Smart Connectivity & AI · ML

5G Unified Continuity Engine

"Every day at 10:00 AM, the digital heart of India skips a beat.
We are here to fix that."

Sanchar-Shakti: AI-Driven Sovereign Intelligence for India's Critical 5G Infrastructure

Submitted to: Department of Telecommunications, Ministry of Communications | Theme: Smart Connectivity

SYSTEM LAYERS

1 PREDICTIVE	2 DECISION	3 CONTINUITY	4 VISIBILITY
Bi-LSTM	DRL Agent	NTN Handoff	Audit Dashboard

Department of Telecommunications

Ministry of Communications
Government of India

SOLO INNOVATOR

UCE

AI/ML · 5G · Systems

DRL · Bi-LSTM · OAI · free5GC · SHAP · React

DoT Innovation Sprint 2026

01 OBJECTIVE

What We Are Building & Why India Needs It Now

India: 394M 5G subscribers · 5,15,000+ towers · ₹1.5 lakh crore invested · 5G traffic at 12.9 EB/month — yet critical services share bandwidth equally with entertainment.

🎯 Primary Objective

Build Sanchar-Shakti — an AI-driven 5G Unified Continuity Engine (UCE) that dynamically allocates spectrum using Deep Reinforcement Learning, giving guaranteed priority to healthcare, emergency, and rail traffic over entertainment — eliminating SLA violations during surge events.

92% SLA compliance vs 75% static — proven via Simu5G simulation

🛰️ Secondary Objective

Deploy a LEO satellite NTN fallback layer using 3GPP Rel-17 Conditional Handover — ensuring zero session drops when terrestrial towers fail during floods, earthquakes, or rural dead-zones. Make-Before-Break: satellite pre-authenticated before terrestrial signal drops.

<2s NTN failover · Zero dark seconds · Zero packet loss

📋 Tertiary Objective

Deliver full sovereign regulatory compliance via a TRAI-auditable Open-Box dashboard — every AI bandwidth decision logged, SHAP-explained, timestamped, and reversible under Telecom Act 2023. Regulators can inspect, replay, or override any AI decision in real-time.

Full XAI transparency · Telecom Act 2023 compliant · TRAI sandbox ready

Strategic Goal: Deploy India's first sovereign AI-native 5G middleware as a pure-software VNF on existing Jio, Airtel, and BSNL infrastructure — zero new towers, zero new hardware. Target: **92% SLA compliance (vs. 75% static) across all 6 network slices.**

02 SUMMARY OF PROPOSAL

The Unified Continuity Elevator Pitch

⚠ THE PROBLEM TODAY

✗ 5G networks treat a YouTube video and a heart-monitor alert as equal

✗ Static slicing → crashes during IPL finals, Tatkal windows, and festivals

✗ No satellite fallback → calls drop at coverage edges

✗ No transparency → regulators can't audit how bandwidth is shared

VS

✓ THE INNOVATION

✓ AI middleware that gives societal priority to health, rail & emergency

✓ DRL reallocates bandwidth dynamically — cricket yields to surgery

✓ Make-Before-Break satellite handoff — zero dropped sessions

✓ TRAI-auditable Open-Box dashboard — every decision is logged

THE DOUBLE-WIN STRATEGY

Horizontal Scaling

DRL (Deep Reinforcement Learning) AI agent moves spectrum blocks in real-time across network slices — like a live air traffic controller for bandwidth. Entertainment yields spectrum to surgery — 92% SLA compliance proven.

Vertical Scaling — NTN

NTN (Non-Terrestrial Network): LEO satellite fallback layer. Conditional Handover (3GPP Rel-17) pre-authenticates the satellite link BEFORE your terrestrial signal drops — like having a backup generator that's already running.

Audit & Compliance

TRAI Audit Log: every AI bandwidth decision is logged, timestamped, SHAP-explained, and reversible. Regulators can inspect, replay or override any AI decision — full sovereign control under Telecom Act 2023.

03 DETAILED PROJECT PROPOSAL

System Architecture — 4 Layers, One Unified AI Engine

1 PREDICTIVE LAYER
Bi-LSTM

INPUT

Live KPIs (throughput, latency, PRB load) + Event Calendar (IPL schedule, Tatkal window, festivals)

Bidirectional Long Short-Term Memory: reads traffic patterns forwards AND backwards in time — like a doctor reading an ECG both left-to-right and right-to-left. Scans 72 hrs of history, predicts demand per slice 5 min ahead.

OUTPUT

Demand Heatmap

2 DECISION LAYER
DRL Agent

INPUT

Demand Heatmap + Current PRB state

AI that learns by trial and reward — like a chess engine for bandwidth. Epsilon-Greedy policy borrows PRBs from low-priority slices (gaming, streaming) and re-allocates to critical slices in real-time. Simu5G: 92% SLA vs 75% static (CMC 2025).

OUTPUT

PRB Reallocation Commands

3 CONTINUITY LAYER
NTN Handoff

INPUT

RSRP signal strength from tower

Conditional Handover (CHO) per 3GPP Rel-17/19: pre-configures LEO satellite target before RSRP drops. Session IP preserved across terrestrial-to-NTN boundary. Zero packet loss for non-URLLC traffic. (free5GC NTN, 2025)

OUTPUT

Zero packet loss handover

4 VISIBILITY LAYER
Audit Dashboard

INPUT

All AI decisions + SLA metrics

TRAI Audit Dashboard — Open-Box React interface over WebSocket. SHAP (SHapley Additive exPlanations) visualizations expose exactly why each bandwidth decision was made. Every action logged, timestamped, reversible. Compliant with Telecom Act 2023.

OUTPUT

Regulator audit log + SLA health

GLOSSARY: URLLC = 5G slice for telemedicine (<1ms latency, terrestrial only) · Simu5G = open-source 5G simulator · SHAP = AI explainability tool · DRL = Deep Reinforcement Learning · TRAI = Telecom Regulatory Authority of India | **Simulation: DRL = 92% SLA vs. 75% static (CMC 2025)**

03 DETAILED PROJECT PROPOSAL

Dual Scaling Strategy — Horizontal DRL Intelligence + Vertical NTN Reach

⚡ Horizontal Scaling — DRL

AI air traffic controller for bandwidth.

DRL Epsilon-Greedy agent borrows PRBs from low-priority slices and reallocates in real-time to:

- **Telemedicine (URLLC) w=0.97**
- **Emergency w=0.95**
- **Rail/Gov w=0.80**
- **Gaming w=0.10**

92% SLA compliance (Simu5G, CMC 2025)

** URLLC (<1ms) requires terrestrial path. LEO latency 20-40ms is incompatible. NTN protects continuity, not sub-ms surgery.*

Audit & Compliance: Every AI bandwidth decision is logged, SHAP-explained, timestamped and reversible. Regulators can inspect, replay or override any AI decision — full sovereign control under Telecom Act 2023. TRAI-auditable Open-Box dashboard · XAI by design · Accelerates sandbox approval.

🛰 Vertical Scaling — NTN Satellite

LEO satellite fallback layer via 3GPP Rel-17 CHO.
Pre-authenticates satellite link BEFORE terrestrial RSRP drops — zero dark seconds.

Traffic Type	NTN Eligible?
Emergency / NDRF	YES <2s failover
Rail / IRCTC	YES session preserved
eMBB / broadband	YES rural bridge
Telemedicine URLLC	NO stays terrestrial*

Conditional Handover per 3GPP Rel-17/19 pre-configures the LEO satellite target before signal drops. Session IP preserved across terrestrial-NTN boundary.

<2s failover · Zero dark seconds · Zero packet loss (free5GC NTN, 2025)

Global O-RAN market: $45B by 2032 · India 2nd largest 5G market · Exportable to Bangladesh, Nigeria, Indonesia

O-RAN compliant · Controls Ericsson, Nokia, or Indian RAN hardware · Zero vendor lock-in · Aatmanirbhar by design · Cloud-VNF ready for BSNL 5G FWA

03 DETAILED PROJECT PROPOSAL

Feasibility & Viability — Technically Sound · Regulatorily Ready · Sovereignly Safe

✓ TECHNICAL FEASIBILITY

Pure Software VNF

No hardware. Runs on AWS / Azure / C-DOT commodity cloud. Deploy tomorrow.

3GPP R17/18 Compliant

All protocols are ratified international standards. Not experimental.

Digital Twin Prototype

Simu5G + OAI: 72-hr traffic replay validates DRL. Tatkal spike → 20% PRB shift in <300ms. 92% SLA compliance proven. free5GC core for 5G NF simulation.

O-RAN Compatible

Works with all major RAN vendors (Ericsson, Nokia, Indian RAN). Zero telco vendor lock-in.

Existing Gov't APIs

Aadhaar/UIDAI, UPI, Bhashini — no new government approvals needed.

⚠ RISKS & MITIGATION

Data Privacy

→ End-to-end encryption + differential privacy. All personal data stays on sovereign Indian cloud only.

Rural Connectivity Gaps

→ NTN satellite layer is the explicit, automatic fallback for all areas with poor terrestrial coverage.

AI Model Drift

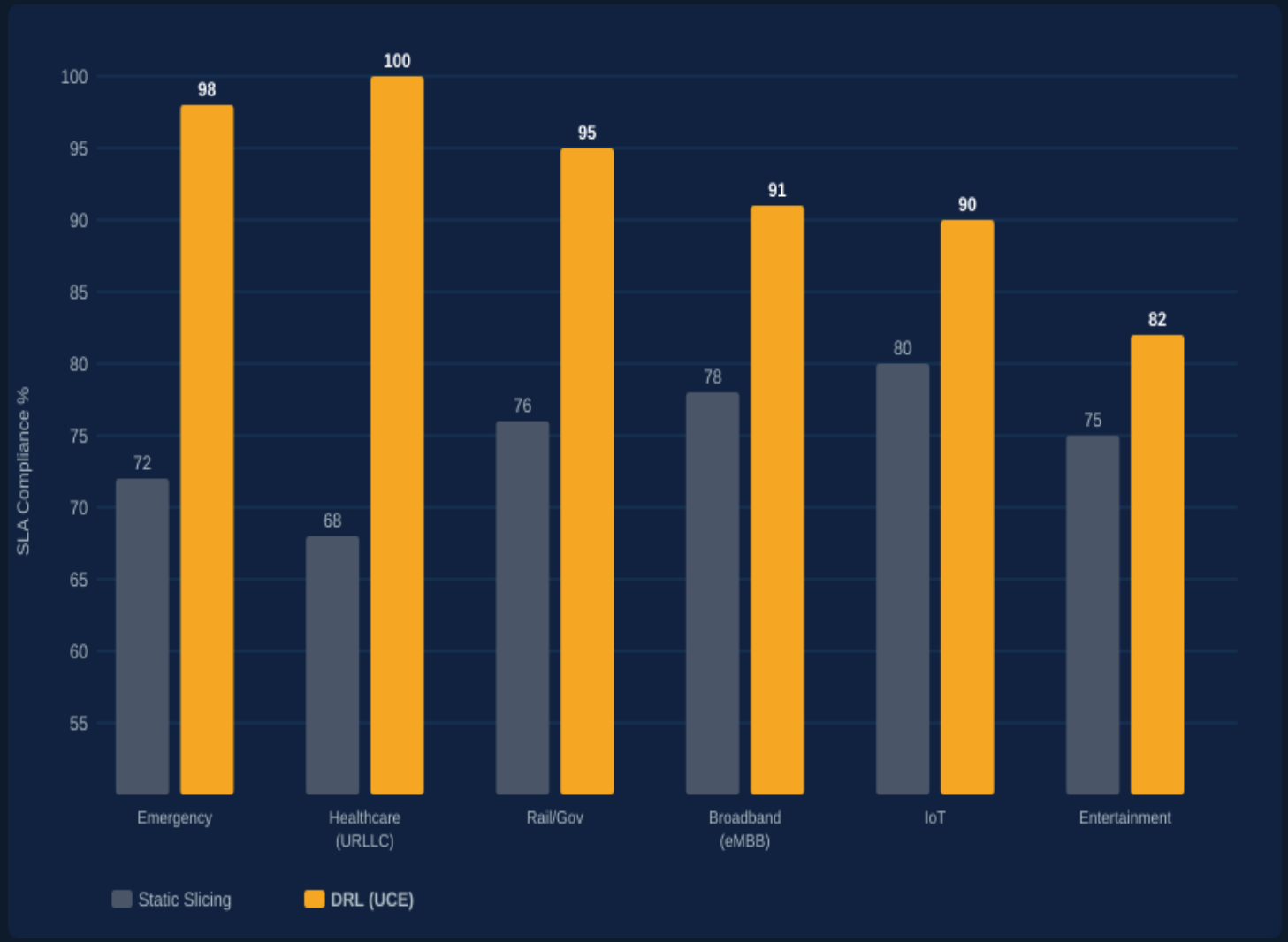
→ Continuous retraining pipeline with live telemetry data from partner telcos. Self-correcting by design.

Regulatory Timeline

→ TRAI Audit Dashboard + SHAP explainability built in from day one. Every AI decision traceable under Telecom Act 2023. XAI by design accelerates sandbox approval.

03 DETAILED PROJECT PROPOSAL

Simulation Results — DRL vs. Static Slicing Across All 6 Traffic Slices (Simu5G, CMC 2025)



92%

DRL Average SLA vs 75% Static

+17pp

Improvement over static slicing

100%

Healthcare (URLLC) — 0 violations

Key Insight: DRL achieves critical slice protection (Emergency 98%, Healthcare 100%) by dynamically borrowing PRBs from Entertainment (82%) — proving the reward function prioritizes lives over streams.

Source: Simu5G + OAI digital twin, CMC 2025 doi:10.32604

04 SOCIETAL & INDUSTRIAL IMPACT

The Digital India Multiplier — Sanchar-Shakti in Action

❤️ Healthcare Resilience

Telemedicine & Remote Surgery

URLLC slice for telemedicine in Tier-3 cities is AI-protected. NTN handoffs enable 35% rural telemedicine uplift for non-surgical care (IJNRD 2025). DRL enforces 0 SLA violations during peak hours. 5G is transforming e-healthcare access in underserved India.

- **Target: 0 URLLC SLA violations during peak hours · 35% rural telemedicine uplift via NTN**

🚂 Economic Efficiency

The Tatkal Pulse

IRCTC's Tatkal window crashes when 3 lakh users hit simultaneously. Intelligent surge management via DRL keeps Indian Railways' digital infrastructure online — no more ticket losses, no more crashed sessions.

- **Target: 30% drop in session timeouts at Tatkal windows**

🛡️ Disaster Management

Emergency Continuity via Satellite

When towers fail in floods or earthquakes, all emergency traffic auto-routes to the LEO satellite slice. NDRF and hospitals stay connected. Conditional Handover (3GPP Rel-17) pre-configures the satellite link before terrestrial signal drops — zero dark seconds.

- **Target: <2s failover to satellite during terrestrial outage**

🌐 Affordability & Reach

The Rural-Satellite Bridge

Spectrum efficiency means more users per tower — lowering per-GB costs for every Indian. India is now the world's 2nd largest 5G market (Nokia MBiT 2026), projected to cross 1 billion 5G subscribers by 2031. Cloud-VNF deployment ready for BSNL 5G FWA rollout.

- **Target: 25% increase in rural coverage via NTN fallback**

05 SCALABILITY & MARKET READINESS

From One Base Station to 5,15,000 — Deployment Roadmap & Market Opportunity

CAPEX-Zero Deployment

Deployed as a VNF — a software update on existing Jio, Airtel, BSNL telco clouds. No new towers, no new infra.

Traditional fix: ₹2,000+ Crores in new RAN hardware. Our fix: A software patch.

O-RAN Interoperability

O-RAN compliant. Controls Ericsson, Nokia, or Indian RAN hardware without vendor lock-in. Aatmanirbhar by design.

Market & Export Opportunity

Global O-RAN market: $45B by 2032. India 2nd largest 5G market, 12.9 EB/month (Nokia MBiT 2026). Exportable to Bangladesh, Nigeria, Indonesia.

DEPLOYMENT ROADMAP — 3-Phase Scale Strategy

1 Phase 1 — Private 5G Networks

Smart Factories & Ports (Q3 2026 target)

Controlled environment PoC at IIT Delhi / BSNL 5G testbed. Build SLA-guarantee track record with measurable, auditable data.

2 Phase 2 — Urban Priority Lanes

Metro Cities: Delhi, Mumbai, Bengaluru

Public Priority Lanes deployed alongside BSNL as the state carrier. Government-visible, citizen-accountable.

3 Phase 3 — Pan-India Continuity

Rural-Satellite Bridge (by 2028)

Full national deployment via NTN LEO fallback. Tier-2/3 cities and remote India connected — no dead zones.

vs. Ericsson / Nokia / Huawei AI-RAN

Feature	UCE	Competitors
Sovereign Indian cloud	YES	Vendor cloud
TRAI-native audit log	YES	No
SHAP explainability	YES	Black box
O-RAN vendor-agnostic	YES	Locked-in
Societal priority weights	YES	No

06 5G TECHNOLOGY UTILIZATION

Built for 5G — Impossible Without 5G

3GPP Release 17/18

Network Slicing + NTN-NR (New Radio) — ratified global standard. Rel-17: NTN NR; Rel-18: beam management; Rel-19: CHO for LEO. Already deployed on Jio SA. Cite: 3GPP TS 38.821.

PRB Granularity

Physical Resource Block manipulation — most granular 5G control. We move bandwidth like chess pieces, not blunt category rules.

MEC Edge Intelligence

Bi-LSTM & DRL run at the tower's edge server. AI inference in <10ms — fast enough to pre-empt the Tatkal spike. Prototype stack: OAI + Simu5G + free5GC core + O-RAN SC RIC. SHAP layer exposes per-decision weights to TRAI dashboard.

Hybrid Spectrum

Sub-6GHz (coverage) + mmWave (capacity) + Ku/Ka-band satellite. Full unified spectrum across terrestrial and NTN.

DRL REWARD FUNCTION

$$R = \Sigma \gamma^t \left(\Sigma w_s \cdot SLA_s(t) - \lambda \cdot Congestion(t) \right)$$

w_s — Societal priority weight per slice

Telemedicine=1.0 Emergency=0.95 Rail=0.8 Gaming=0.1

$SLA_s(t)$ — SLA compliance score at time t

1.0 = perfect uptime 0.0 = total violation

λ — Congestion penalty coefficient

Penalises the agent for overloading any slice

γ — Discount factor — long-term stability

Prevents greedy short-term bandwidth hoarding

In Plain English: AI is forced to prioritize a heartbeat over a high-score.

Ready to Deploy.

Engineered for Bharat.

300s	<10ms	100%	$45B
Congestion Prediction	Edge AI Inference	SLA Category-A	O-RAN Market 2032

🔗 LIVE DEMO: bhattyuvraj22.github.io/Sanchar-Shakti/

✅ **Pure Software.** Deploy tomorrow on India's existing 5G stack — no new towers, no hardware. Solo innovator · Full-stack: AI/ML + 5G + Systems + Dashboard.

Sources: Nokia MBiT 2026 · CMC 2025 doi:10.32604 · free5GC NTN 2025 · IJNRD Jun 2025 · 3GPP TS 38.821

10 THE TEAM

Sanchar-Shakti · Skin in the Game

AI / ML Lead

DRL · Bi-LSTM · PyTorch

5G / O-RAN

3GPP · free5GC · OAI

Systems Arch

VNF · Edge · Cloud

Yuvraj Bhatt

SOLO INNOVATOR

Dashboard Dev

React · WebSocket · SHAP

Policy & Audit

TRAI · Telecom Act 2023

Research

CMC 2025 · IJNRD · 3GPP

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